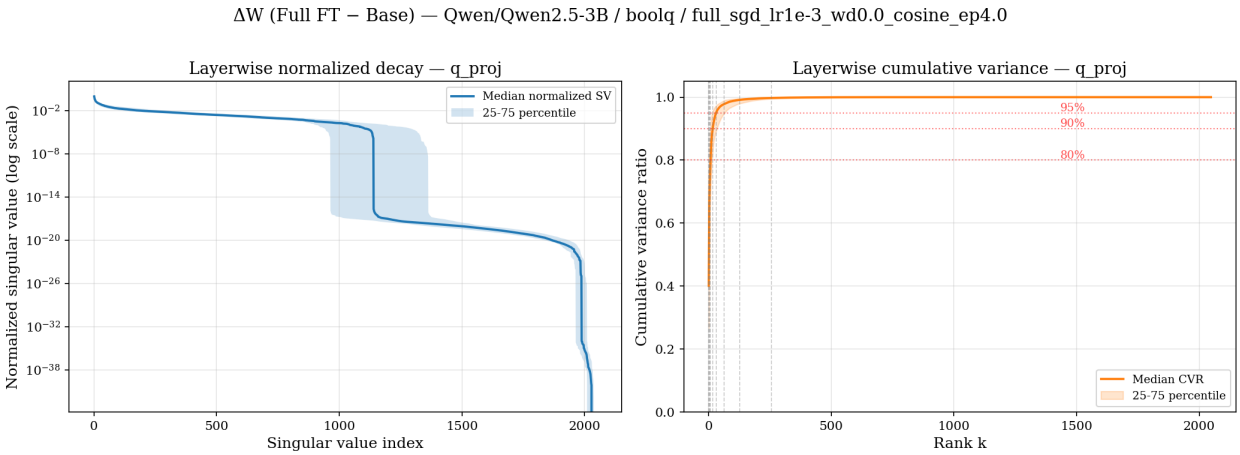
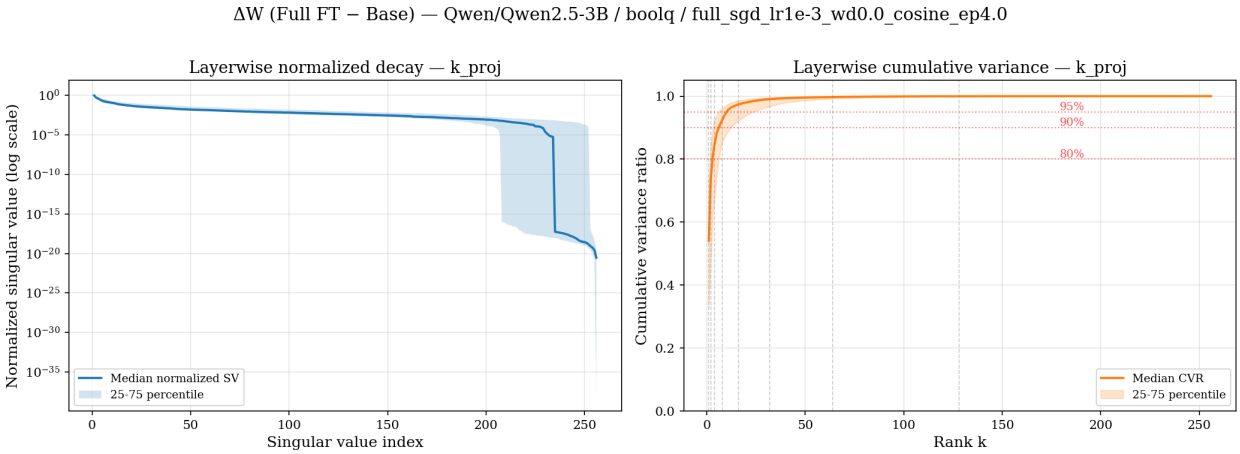
Table 1: Full fine-tuning results on Qwen2.5-3B with SGD.

Dataset	LR	Epochs	Batch	Accum	Momentum	Scheduler	Warmup	Acc (norm)
ARC-Easy	1e-4	4	4	4	0.9	cosine	0.2	80.5
ARC-Challenge	1e-4	4	4	4	0.9	cosine	0.2	50.0
BoolQ	1e-3	4	4	4	0.9	cosine	0.2	86.5

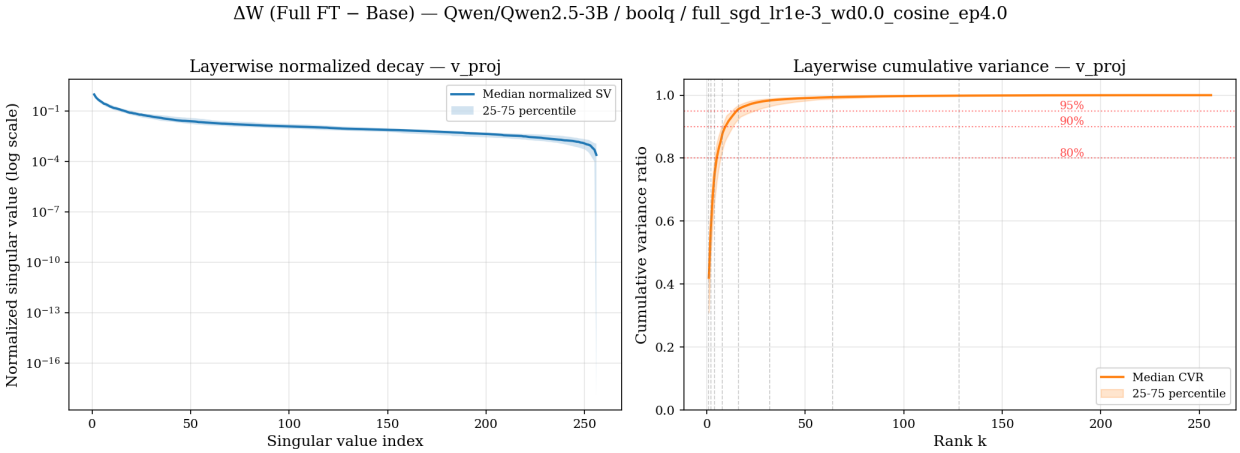
Figure 1: SVD of ΔW (Full FT − Base) for Q, K, V projections — BoolQ (lr=1e-3).



(a) Q projection

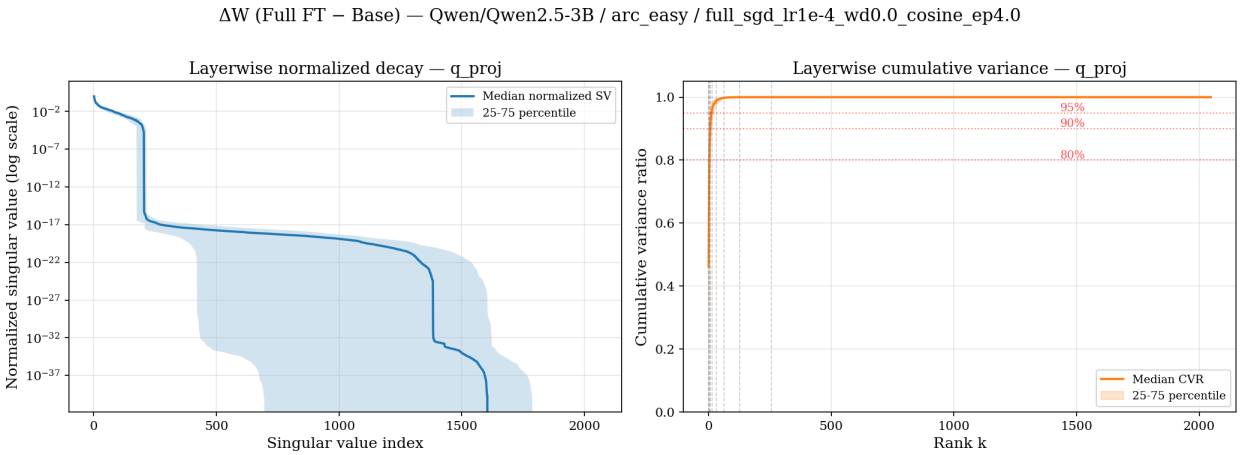


(b) K projection

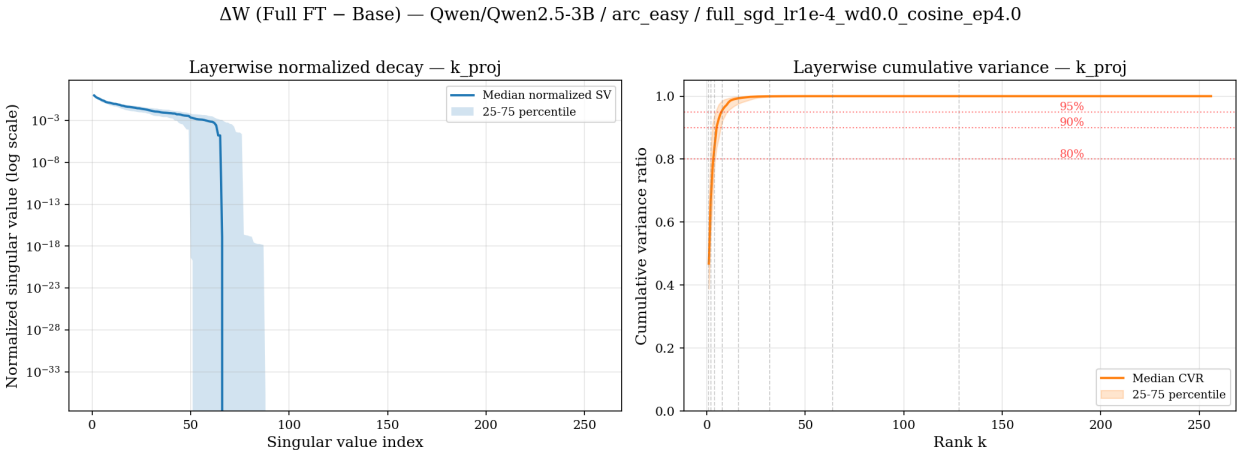


(c) V projection

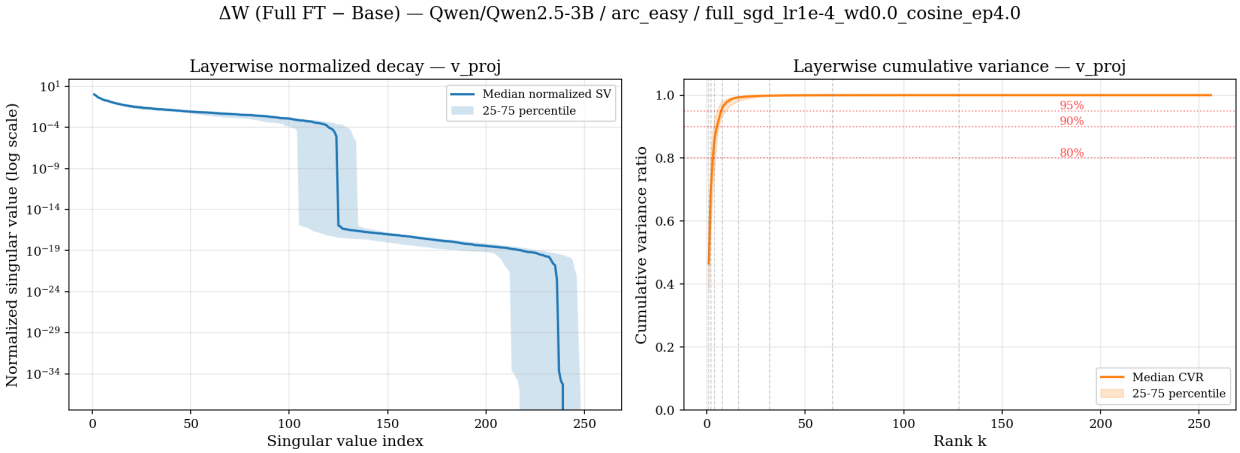
Figure 2: SVD of ΔW (Full FT − Base) for Q, K, V projections — ARC-Easy.



(a) Q projection

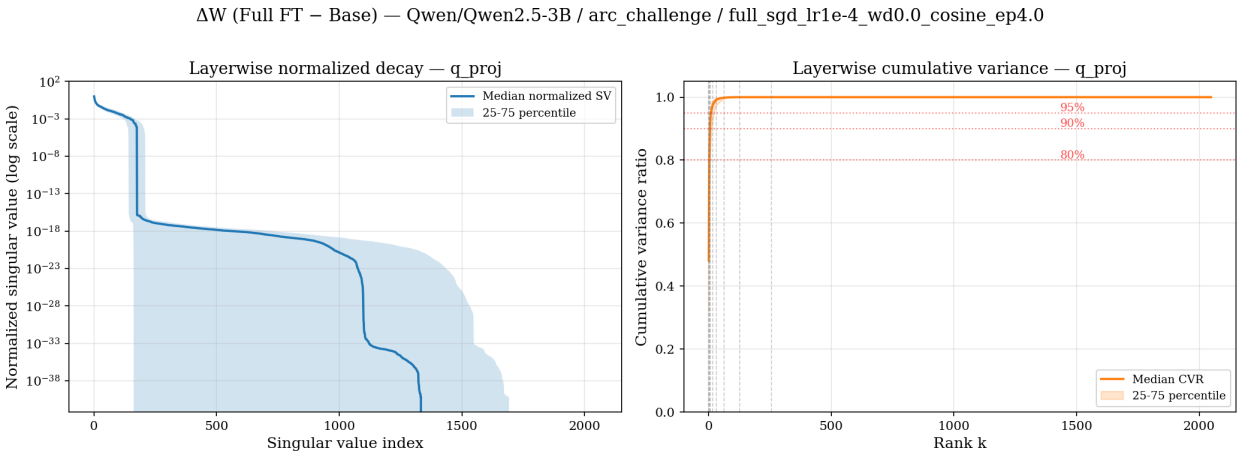


(b) K projection

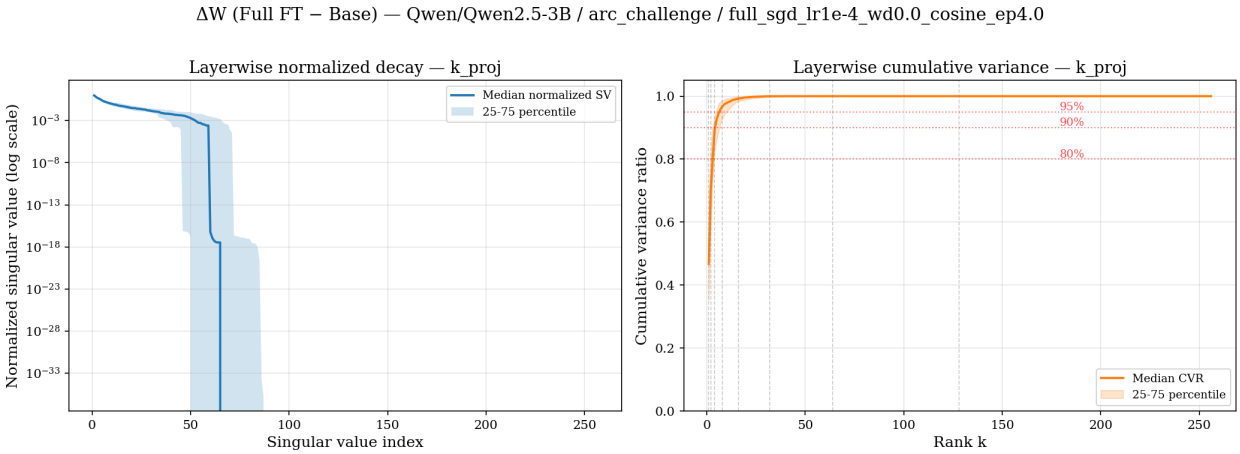


(c) V projection

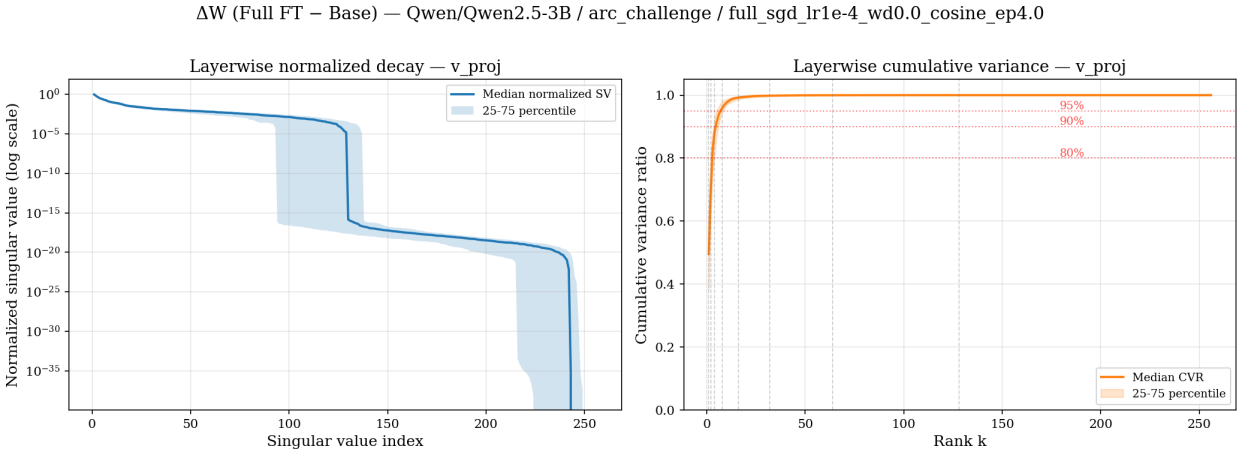
Figure 3: SVD of ΔW (Full FT − Base) for Q, K, V projections — ARC-Challenge.



(a) Q projection



(b) K projection



(c) V projection